



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Modern Physics : Course Introduction

Modern Physics Instructors
Department of Physics,
Indian Institute of Technology Hyderabad.

3rd Sep 2026

Course Instructors



Dr. Divya Sachdeva
Office: PH211
Email: divyasachdeva@phy.iith.ac.in



Prof. Vandana Sharma
Office: PH403
Email: vsharma@phy.iith.ac.in



Dr. Archak Purkayastha
Office: PH314
Email: archak.p@phy.iith.ac.in



Dr. Saurabh Sandilya
Office: PH309
Email: saurabh@phy.iith.ac.in

Schedule

Monday	Tuesday	Wednesday	Thursday	Friday
31-Aug-2026	01-Sep-2026	02-Sep-2026	03-Sep-2026	04-Sep-2026
A 9:00 Intro (All)		A 11:00 V1	A 10:00 D1	
B 10:00 Intro (All)		B 9:00 V1	B 11:00 D1	
07-Sep-2026	08-Sep-2026	09-Sep-2026	10-Sep-2026	11-Sep-2026
A 9:00 V2		A 11:00 V3	A 10:00 D2	
B 10:00 V2		B 9:00 V3	B 11:00 D2	
14-Sep-2026	15-Sep-2026	16-Sep-2026	17-Sep-2026	18-Sep-2026
Holiday: Ganesh Chaturthi		A 11:00 V4	A 10:00 D3	
		B 9:00 V4	B 11:00 D3	
21-Sep-2026	22-Sep-2026	23-Sep-2026	24-Sep-2026	25-Sep-2026
A 9:00 V5		A 11:00 V6	A 10:00 D4	
B 10:00 V5		B 9:00 V6	B 11:00 D4	
28-Sep-2026	29-Sep-2026	30-Sep-2026	01-Oct-2026	02-Oct-2026
A 9:00 D5		A 11:00 D6	A 10:00 Buffer	Holiday: Gandhi Jayanti
B 10:00 D5		B 9:00 D6	B 11:00 Buffer	
05-Oct-2026	06-Oct-2026	07-Oct-2026	08-Oct-2026	09-Oct-2026
Mid-Sem		A 11:00 A1	A 10:00 S1	
Mid-Sem		B 9:00 A1	B 11:00 S1	
12-Oct-2026	13-Oct-2026	14-Oct-2026	15-Oct-2026	16-Oct-2026
Mid-Sem Paper Show		A 11:00 A2	A 10:00 S2	
Mid-Sem Paper Show		B 9:00 A2	B 11:00 S2	
19-Oct-2026	20-Oct-2026	21-Oct-2026	22-Oct-2026	23-Oct-2026
Semester Break	Semester Break	Semester Break	Semester Break	Semester Break
26-Oct-2026	27-Oct-2026	28-Oct-2026	29-Oct-2026	30-Oct-2026
A 9:00 A3		A 11:00 A4	A 10:00 S3	
A 10:00 A3		B 9:00 A4	B 11:00 S3	
02-Nov-2026	03-Nov-2026	04-Nov-2026	05-Nov-2026	06-Nov-2026
A 9:00 A5		A 9:00 A6	A 10:00 S4	
A 10:00 A5		A 10:00 A6	B 11:00 S4	
09-Nov-2026	10-Nov-2026	11-Nov-2026	12-Nov-2026	13-Nov-2026
A 9:00 S5		A 11:00 S6	A 10:00 Buffer	
A 10:00 S5		B 9:00 S6	B 11:00 Buffer	
16-Nov-2026	17-Nov-2026	18-Nov-2026	19-Nov-2026	20-Nov-2026
End-Sem	Exam Week	End-Sem Paper Show	Exam Week	Exam Week
End-Sem		End-Sem Paper Show		

Syllabus

Dr. Divya Sachdeva

- Electromagnetic Waves
- Photoelectric Effect
- X-ray, X-ray diffraction
- De Broglie Waves
- Uncertainty Principle
- Electron Spin
- Fermions and Bosons
- Spin-orbit coupling
- X-ray Spectra

Prof. Vandana Sharma

- **Lecture 1:** Atomic Structure, Atomic Spectra
- **Lecture 2:** Bohr's Atom and Energy levels
- **Lecture 3:** Quantum Mechanics—Operators, Particle in a Box
- **Lecture 4:** Oscillator, Tunnelling Effect
- **Lecture 5:** Hydrogen atom and Quantum Numbers
- **Lecture 6:** Zeeman Effect, Selection Rules

Dr. Archak Purkayastha

- The concept of ensemble in Statistical Physics.
- Gas of distinguishable and indistinguishable particles: Maxwell-Boltzmann distribution, Bose-Einstein distribution, Fermi distribution.
- Examples (if time permits): Planck's radiation law, specific heat of solids, free electrons in a metal.

Dr. Saurabh Sandilya

- **Lecture 1 :** Radioactivity
- **Lecture 2 :** Radiation units
- **Lecture 3:** Discovery of Nucleus and Rutherford scattering
- **Lecture 4:** Nuclear radii
- **Lecture 5:** Binding Energy and Stable Nuclei Liquid Drop Model of Nuclei
- **Lecture 6:** Particle Physics : Fundamental Particles and forces.

Sr. nr	Name	Email
1	PARISHKRITH	PH25RESCH12001@iith.ac.in
2	SIVA MAJJI	PH25RESCH11010@iith.ac.in
3	LILABATI TANDI	PH25RESCH01004@iith.ac.in
4	PRIYARANJAN KUMAR JHA	PH24RESCH12002@iith.ac.in
5	SAURABH SIHAG	PH24RESCH11016@iith.ac.in
6	RUDRESH KUNDUGOLMATH	PH24RESCH11013@iith.ac.in
7	ABHIRUP CHATTERJEE	PH24RESCH11005@iith.ac.in
8	JAIDEEP NIRMAL	EP23BTECH11013@iith.ac.in
9	AVANTIKA DHANANJAY	EP23BTECH11004@iith.ac.in
10	SAVIREDDY ANSIKA	EP24BTECH11028@iith.ac.in
11	JAIDHAR RAMAGIRI	EP23BTECH11021@iith.ac.in
12	ARAV P	EP23BTECH11003@iith.ac.in
13	ARPIT SURYAVANSHI	EP24BTECH11005@iith.ac.in
14	VISHAL	EP24BTECH11033@iith.ac.in
15	AMIR HUSAIN	PH25MTECH11008@iith.ac.in
16	SURIYANARAYANAN V G	PH25MTECH11007@iith.ac.in
17	UPAM CHOUDHURY BRAHMA	PH25MTECH11005@iith.ac.in
18	DEVESH KUMAR SINGH	PH25MTECH11002@iith.ac.in
19	AJAY KUMAR SAHANI	PH25MTECH11001@iith.ac.in



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

**Thank
You!**